

From geometric projection to posterior covariance bundles

Main idea. Replace “projection onto a manifold” by a *bundle of posterior covariances*. Let

$$Y_t = X + \sqrt{t} Z, \quad m_t(y) := \mathbb{E}[X \mid Y_t = y], \quad A_t(y) := \frac{1}{t} \text{Cov}(X \mid Y_t = y).$$

The scale-dependent target is not a set M_t , but the weighted object

$$\mathfrak{F}_t := \text{Law}(Y_t, m_t(Y_t), A_t(Y_t)).$$

Here $m_t(y)$ is a distributional projection of y , and $A_t(y)$ is its soft tangent tensor.

Exact identities. If p_t is the density of Y_t , then

$$s_t(y) := \nabla \log p_t(y) = \frac{m_t(y) - y}{t}, \quad Dm_t(y) = A_t(y), \quad Ds_t(y) = \frac{A_t(y) - I}{t}.$$

Thus the score points *exactly* toward the posterior barycentre. Its local geometry is encoded by the spectrum and spatial variation of A_t .

Small-noise manifold picture. If X has density ρ on a smooth d -manifold M , then, formally,

$$A_t(x) \simeq P_{T_x M}, \quad m_t(x) - x \simeq t \nabla_M \log \rho(x) + \frac{t}{2} H_M(x).$$

Consequently, for $h \in N_x M$,

$$s_t(x + h) \simeq \nabla_M \log \rho(x) + \frac{1}{2} H_M(x) - \frac{h}{t}.$$

This is the quantitative content of “the score projects onto M ”: $Ds_t \simeq -P_N/t$. The tangential part transports density; the normal part contracts toward the support.

All-scale local expansion and hard ridge. At a d -ridge point r , let N_r denote the spectral normal space and $A_{N,t} = P_{N_r} A_t(r) P_{N_r}$. For $h \in N_r$,

$$s_t(r + h) = P_{T_r} s_t(r) - \frac{I - A_{N,t}}{t} h + O\left(\frac{\|\kappa_{3,t}\| \|h\|^2}{t^3}\right),$$

where $\kappa_{3,t}$ is a local posterior third cumulant. The classical ridge is only the thresholded shadow

$$P_{N_r}(m_t(r) - r) = 0, \quad a_{d+1}(r) < 1,$$

where a_i are the eigenvalues of $A_t(r)$. Hence

$$a_i \simeq 0 : \text{strong normal projection}, \quad a_i \simeq 1 : \text{soft tangent direction}, \quad a_i > 1 : \text{ambiguity/local expansion}.$$

The hard ridge bifurcates when $a_{d+1} = 1$ or the spectral gap closes, although the posterior object (m_t, A_t) remains smooth.

Diffusion ODE. Along a decreasing noise schedule t_τ , the reverse probability-flow ODE has the form

$$\dot{y}_\tau = \beta_\tau(m_t(y_\tau) - y_\tau), \quad D\dot{y}_\tau = -\beta_\tau(I - A_t).$$

It is a relaxation toward the posterior barycentre; the rates $\beta_\tau(1 - a_i)$ quantify contraction, stiffness, and transient expansion direction by direction.

Complexity across scales. Three useful quantities are

$$d_{\text{eff}}(t) := \mathbb{E} \text{tr} A_t = \frac{\text{mmse}(X \mid Y_t)}{t}, \quad \Delta_{\text{proj}}(t) := \mathbb{E} \sum_i \min\{a_i^2, (a_i - 1)^2\}, \quad \mathcal{A}(t) := \mathbb{E} \sum_i (a_i - 1)_+^2.$$

They measure effective dimension, deviation from a manifold-like projector, and ambiguous expansive directions. If $S(t) := \text{tr} \text{Cov}(m_t(Y_t))$, then the exact evolution laws are

$$S'(t) = -\mathbb{E} \|A_t\|_{\mathbb{F}}^2, \quad t d'_{\text{eff}}(t) = \mathbb{E} \text{tr}(A_t^2 - A_t).$$

Thus the barycentric skeleton loses spatial complexity monotonically, whereas effective dimension can temporarily increase precisely when posterior ambiguity creates eigenvalues $a_i > 1$.